HireGen AI

Powering the Future of Intelligent Hiring

BUSINESS REQUIREMENTS DOCUMENT

AI-Powered Recruitment and Candidate Assessment Platform

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1. Executive Summary

HireGen AI is an enterprise-grade, AI-powered recruitment and candidate assessment platform designed to transform the end-to-end hiring lifecycle. The platform is built on a mature, production-quality Node.js/Express/MongoDB foundation and is being strategically enhanced with state-of-the-art Generative AI, semantic search, explainability layers, and real-time analytics capabilities.

The recruitment industry is undergoing rapid transformation. Manual shortlisting, subjective evaluation, and keyword-based ATS systems are increasingly inadequate in a competitive talent market. Research indicates that AI-driven platforms can reduce manual shortlisting effort by up to 97%, improve candidate search relevance by 38%, and increase interview-to-offer conversion rates by 30–45%. HireGen AI is positioned to capture this opportunity by delivering an intelligent, explainable, and bias-free hiring experience.

This Business Requirements Document (BRD) defines the complete set of business objectives, functional requirements, non-functional requirements, integration specifications, compliance considerations, and a phased implementation roadmap for the HireGen AI platform. It serves as the authoritative reference for all product, engineering, and stakeholder alignment activities.

2. Project Overview

2.1 Project Background

The current HireGen AI platform is a fully operational, full-stack recruitment application comprising 10 MongoDB schemas, 8 controllers, and 9 service modules. It supports end-to-end recruitment workflows including resume ingestion, candidate communication, and role-based access control. Basic AI capabilities such as MCQ generation and behavioural assessments have been implemented, establishing a foundation for more advanced AI-driven features.

The strategic initiative described in this document aims to evolve HireGen AI from a functional Applicant Tracking System (ATS) into an intelligent, production-grade recruitment intelligence engine — leveraging Large Language Models (LLMs), vector embeddings, semantic search, and explainable AI to deliver measurable improvements in recruiter productivity and hiring quality.

2.2 Scope

In-Scope

- AI-based candidate scoring, ranking, and explainability (XAI)
- Semantic and vector search using RAG (Retrieval-Augmented Generation) pipelines
- AI-generated candidate summaries and recruiter chat assistant
- Automated interview question generation and feedback analysis
- Resume fraud, risk, and authenticity detection
- AI-powered job description (JD) and email content generation
- Enhanced analytics dashboards and salary benchmarking
- LLM gateway with multi-model routing, circuit breakers, and fallbacks
- Microservices architecture migration with Docker/Kubernetes orchestration

- Monitoring and observability integration (CloudWatch / Grafana)
- Retention and enhancement of all existing platform workflows

Out-of-Scope

- Mobile application development (iOS/Android native apps)
- Video interview platform integration (e.g., Zoom, Microsoft Teams native modules)
- Background check third-party API integrations (e.g., HireRight, Checkr) — Phase 5+
- Payroll or HRIS system integrations
- Candidate-facing career portal redesign
- GDPR, SOC2, and EEOC compliance certification (recommended for future phases)

3. Business Objectives & Success Criteria

The following table defines the strategic business objectives for HireGen AI, along with measurable success criteria and target KPIs that will be used to evaluate platform performance post-implementation.

#	Business Objective	Success Criteria / KPI	Target
BO-01	Automate candidate shortlisting to reduce recruiter manual effort	Reduction in manual shortlisting time	≥ 97%
BO-02	Improve candidate-to-job relevance through semantic matching	Search relevance improvement (NDCG)	≥ 38%
BO-03	Increase quality of hire by surfacing best-fit candidates	Interview-to-offer conversion rate improvement	30–45%
BO-04	Build recruiter trust through explainable, transparent AI decisions	XAI justification coverage on all scored candidates	100%
BO-05	Reduce bias in the hiring process through AI-assisted JD generation	Bias-flagged terms removed from JDs	Measurable reduction
BO-06	Protect hiring integrity through fraud and authenticity detection	AI-generated / fraudulent resumes flagged	≥ 90% precision
BO-07	Ensure platform availability and AI service resiliency	System uptime and LLM fallback success rate	≥ 99.5% uptime
BO-08	Enable data-driven hiring decisions through real-time dashboards	KPI dashboard adoption by HR Admins and Managers	100% rollout

4. Stakeholders & User Roles

HireGen AI operates with a well-defined role hierarchy enforced through JWT authentication and Role-Based Access Control (RBAC). The following table outlines each stakeholder group, their responsibilities, platform interactions, and access permissions.

Role	Description	Key Responsibilities	Platform Interactions	Access Level
Recruiter	Primary day-to-day user of the platform	Source, evaluate, and manage candidates; conduct semantic search; review AI Fit Scores	Candidate pipeline, scoring, search, interview tools, email automation	Full — Read/Write on candidates, jobs, assessments
HR Admin	Administrative manager of recruitment operations	Manage job postings, standardise JDs, oversee hiring consistency, review analytics	JD Generator, dashboards, email templates, user management	Full — Read/Write/Admin on all HR content
Hiring Manager	Business stakeholder driving hiring decisions	Review shortlists, evaluate AI Fit Scores and summaries, make final hiring decisions	Candidate summaries, ranked shortlists, interview feedback reports	Read — Candidate profiles and reports
Interviewer	Technical or domain expert conducting candidate interviews	Conduct interviews using AI-generated questions; submit structured feedback	Interview question generator, feedback submission forms	Restricted — Assigned candidates only
Candidate / Applicant	Individual applying for a position	Submit resume, complete MCQ/behavioural assessments, receive automated communications	Application portal, assessment module, email communications	Self-service — Own profile and assessments only
Auditor	Compliance and governance reviewer	Review XAI explanations for hiring decisions, audit logs, and workflow history	Audit log viewer, XAI reports, compliance dashboards	Read-only — Audit and compliance data
Executive / Management	Senior leadership overseeing talent acquisition strategy	Monitor recruitment KPIs, pipeline health, and strategic hiring metrics	Executive analytics dashboard, KPI summaries	Read-only — Aggregated analytics

5. Current System Overview

5.1 As-Is Architecture

The existing HireGen AI platform is a monolithic, full-stack web application built on the MERN-adjacent stack (Node.js, Express.js, MongoDB, EJS). It is a production-quality system with the following architectural components:

Component	Technology / Module	Purpose
Backend Framework	Node.js + Express.js	REST API layer; 8 controllers, 9 service modules
Database	MongoDB (10 schemas)	Candidate, job, assessment, audit, and analytics data storage
Frontend/Templating	EJS Templates	Server-side rendered views for recruiter and admin interfaces
Resume Parsing	pdf-parse	Text extraction from PDF resume uploads
File Storage	AWS S3	Secure cloud storage for resumes and media assets
Email Automation	Nodemailer	Transactional email delivery for candidate communications
Authentication	JWT + RBAC	Secure session management and role-based data access
File Upload	Multer	Multipart/form-data handling for resume and asset uploads
AI Layer	GROQ API	LLM inference for MCQ generation and behavioural assessments
Analytics	MongoDB Aggregation Pipelines	KPI computation, reporting, and dashboard data

5.2 Current Limitations

- Keyword-only resume matching: no semantic understanding of candidate skills or role context
- Black-box AI outputs: MCQ and basic scoring lack explainability or justification
- Monolithic architecture: no independent scaling of AI-heavy services (parsing, scoring)
- No vector search capability: prevents advanced semantic candidate-to-job matching
- No fraud or resume authenticity detection mechanisms
- No salary benchmarking or market-based compensation intelligence
- No AI-generated JDs or structured email communication templates
- Limited NLP: pdf-parse extracts raw text but lacks entity recognition (names, titles, skills)
- No LLM resiliency: single GROQ API dependency with no fallback routing or circuit breaking
- No interview intelligence: no AI-generated questions or structured feedback analysis

6. Proposed Solution Architecture

HireGen AI will transition from a monolithic application to a modular, cloud-native microservices architecture. Each service is independently deployable, scalable, and observable. The architecture is designed for high availability, fault tolerance, and AI service resiliency.

- **API Gateway / Service Mesh**

Centralised routing layer for all client requests. Routes traffic to appropriate microservices, enforces rate limiting, and manages authentication tokens.

- **NLP & Resume Parsing Service**

Dedicated Node.js or Python service using pdf-parse and spaCy for advanced text extraction, named entity recognition (NER) for candidate names, titles, skills, and certifications.

- **Vector Embedding Service**

Generates high-dimensional vector representations of resumes and job descriptions using BERT or OpenAI embedding models. Stores vectors in a dedicated vector database (Pinecone / Weaviate / FAISS).

- **AI Scoring & Ranking Engine**

Computes the AI Fit Score using cosine similarity between candidate and job vectors, weighted by custom job parameters (skills, experience, education). Returns ranked candidate shortlists.

- **LLM Gateway Layer**

Central orchestration layer for all generative AI tasks. Implements multi-model routing (GPT-4 → Claude → GPT-3.5 fallback), circuit breakers, request throttling, and automatic retries to ensure continuous availability.

- **Interview Intelligence Service**

Generates tailored interview questions from JD + candidate profile; transcribes and scores interview responses against defined competency rubrics.

- **Content Generation Service**

Handles AI JD generation, email drafting (invitations, rejections, reminders), and candidate summary generation via LLM prompts.

- **Fraud & Risk Detection Service**

Analyses resume text for AI-generated content patterns, tone inconsistencies, job-hopping signals, and expired certification flags. Outputs an authenticity score.

- **Analytics & Reporting Service**

Aggregates pipeline metrics from MongoDB, computes KPIs (time-to-hire, offer acceptance rate, funnel conversion), and feeds real-time dashboards.

- **Monitoring & Observability Layer**

AWS CloudWatch and Grafana integrations for tracking AI call latency, error rates, fallback incidents, and service health. Includes NDCG-based feedback loops for search quality improvement.

7. Functional Requirements

All functional requirements are assigned a unique identifier, priority level, owner user role(s), and clear acceptance criteria. Priority levels: High (P1), Medium (P2), Low (P3).

7.1 AI-Based Candidate Scoring & Ranking

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-001	AI Fit Score Computation	The system shall compute a composite 'AI Fit Score' (0–100) for each candidate-job pair based on skills match, years of experience alignment, and education criteria.	P1	Recruiter, Hiring Manager	Given a candidate profile and a job posting, the system returns an AI Fit Score within 5 seconds with skill, experience, and education sub-scores.
FR-002	Custom Job Weight Configuration	HR Admins shall be able to configure custom scoring weights for skills, experience, and education per job role.	P1	HR Admin	Weight configuration is saved per job; re-scoring a candidate pool reflects the updated weights within 30 seconds.
FR-003	Ranked Candidate Shortlist	The system shall automatically generate a ranked shortlist of candidates, sorted by AI Fit Score in descending order.	P1	Recruiter	A ranked shortlist is generated and displayed on the candidate pipeline view, sortable and filterable by score range.
FR-004	Batch Scoring	The system shall support batch scoring of all applicants for a given job posting simultaneously.	P2	Recruiter, HR Admin	All candidates for a job are scored within 60 seconds for a batch of up to 500 applicants.

7.2 Explainable AI (XAI)

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-005	Score Justification Panel	For every AI Fit Score, the system shall display a justification panel showing matched skills (✓), missing skills (✗), experience delta, and an overall narrative summary.	P1	Recruiter, Hiring Manager, Auditor	Clicking on any candidate score reveals a justification panel with at least 3 scored dimensions and 1 narrative summary sentence.
FR-006	Interview Focus Recommendations	The system shall highlight areas of concern (e.g., skill gaps, certification gaps) as recommended interview focus areas.	P2	Recruiter, Interviewer	Interview focus areas are surfaced as tagged labels on the candidate profile view.
FR-007	Audit-Ready Score Reports	The system shall generate downloadable XAI compliance reports for any candidate-job pair, suitable for recruiter or auditor review.	P2	Auditor, HR Admin	A PDF/CSV XAI report is generated on demand, including score, justification, and timestamp of evaluation.

7.3 Semantic & Vector Search

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-008	Resume & JD Vectorisation	The system shall vectorise all uploaded resumes and active job descriptions	P1	System (Automated)	Within 10 seconds of upload, a resume is converted to a vector and

		into high-dimensional embeddings using BERT or OpenAI embedding models.			stored in the vector database.
FR-009	Cosine Similarity Candidate Search	The system shall support semantic candidate search using cosine similarity against job description vectors via Pinecone, Weaviate, or FAISS.	P1	Recruiter	A semantic search query returns relevant candidates in under 3 seconds, with relevance scores displayed.
FR-010	Natural Language Recruiter Search	Recruiters shall be able to search for candidates using free-text natural language queries (e.g., 'backend engineer with 5+ years cloud experience').	P1	Recruiter	Natural language queries return a ranked list of candidates with semantic relevance scores; irrelevant results are below the top 10.
FR-011	Cross-Role Semantic Matching	The system shall match candidates across role title variations (e.g., 'software architect' matched to 'senior software engineer') based on vector proximity.	P2	Recruiter	Cross-role matching surfaces at least 80% of manually identifiable relevant candidates in test datasets.

7.4 AI-Generated Summaries & Chat Assistant

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria

FR-012	Automated Candidate Profile Summary	The system shall auto-generate a concise bullet-point summary (4–6 points) for each candidate highlighting key skills, certifications, years of experience, and notable attributes.	P1	Recruiter, Hiring Manager	Summary is auto-generated within 5 seconds of candidate profile view; content is derived from the parsed resume.
FR-013	Recruiter Chat Assistant	A conversational AI assistant shall allow recruiters to query candidate data using natural language (e.g., 'Show me candidates with AWS Certified and 5+ years experience').	P2	Recruiter	Chat assistant returns accurate candidate lists in under 4 seconds; supports follow-up clarifications in the same session.

7.5 Interview Automation

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-014	AI Interview Question Generation	The system shall automatically generate a set of tailored technical and behavioural interview questions based on the job description and candidate's specific skill profile.	P1	Recruiter, Interviewer	A set of 8–12 questions (split technical/behavioural) is generated per candidate-job pair within 10 seconds.

FR-015	Interview Response Transcription	The system shall support upload or integration of interview response transcripts for analysis.	P2	Recruiter	Transcript upload is supported in text/PDF format and processed for analysis within 15 seconds.
FR-016	Rubric-Based Feedback Scoring	The system shall score interview responses against pre-defined or AI-generated competency rubrics and output a structured evaluation report.	P2	Recruiter, Hiring Manager	Evaluation report includes an overall score, per-competency breakdown, strengths, weaknesses, and a hiring recommendation (Recommend / Hold / Reject).

7.6 Risk, Fraud & Quality Detection

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-017	Resume Authenticity Scoring	The system shall compute a Resume Authenticity Score (0–100) for each submitted resume, detecting AI-generated content patterns, unusual phrasing, and tone inconsistencies.	P2	Recruiter, HR Admin	An authenticity score and flag status (Authentic / Suspicious / High Risk) is displayed on every candidate profile.
FR-018	Red Flag Detection	The system shall identify and flag patterns indicative of risk: frequent job-hopping (>3 roles in 2 years), unexplained	P2	Recruiter	Risk flags are visually highlighted on the candidate profile with explanatory tooltips.

		employment gaps >6 months, and expired certifications.			
FR-019	Resume-LinkedIn Consistency Check	The system shall support optional cross-verification of resume data against LinkedIn profile data where available.	P3	Recruiter, HR Admin	Inconsistencies between resume and LinkedIn data are flagged with field-level discrepancy annotations.

7.7 AI Content Generation

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-020	AI Job Description Generator	The system shall generate a full, professionally formatted job description from a brief role summary input, with bias-flagging for exclusionary language.	P1	HR Admin, Recruiter	A complete JD (500–800 words) is generated within 15 seconds; bias-flagged terms are highlighted with suggested alternatives.
FR-021	AI Email Communication Generator	The system shall auto-generate tailored recruitment emails (interview invitations, rejections, follow-ups, reminders) using candidate and job context.	P2	Recruiter, HR Admin	Email drafts are generated within 5 seconds; drafts are editable before sending; tone is appropriate to email type.

7.8 Analytics & Dashboards

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-022	Recruitment KPI Dashboard	The system shall provide a real-time recruitment KPI dashboard displaying: open jobs, total applicants, time-to-hire, offer acceptance rate, and pipeline funnel metrics.	P1	HR Admin, Executive	Dashboard loads within 3 seconds; all KPIs are refreshed at least every 15 minutes.
FR-023	Visual Candidate Pipeline	The system shall render a visual, stage-by-stage candidate pipeline (Applied → Screened → Interviewed → Offered → Hired) with counts per stage.	P1	Recruiter, HR Admin	Pipeline view is interactive; clicking a stage filters the candidate list to that stage.
FR-024	Salary Benchmarking	The system shall suggest competitive salary ranges for each job role based on candidate skills, experience, location, and integrated market data.	P2	HR Admin, Hiring Manager	Salary range is surfaced as a widget on the job posting view, sourced from integrated market data APIs.
FR-025	Skill Demand Trend Charts	The system shall display charts of top in-demand skills across active job postings to	P3	HR Admin, Executive	Trend chart is rendered as a bar/line chart; updated weekly from active job

		inform sourcing strategy.			descriptions.

7.9 Retained Existing Workflows

ID	Title	Description	Priority	User Role(s)	Acceptance Criteria
FR-026	Resume PDF Parsing (Retained)	The platform shall retain and enhance resume parsing via pdf-parse, augmented with spaCy NER for structured extraction of names, titles, and skills.	P1	System	Parsed fields (name, email, skills, education, experience) are extracted with ≥95% accuracy on standard PDF formats.
FR-027	MCQ & Behavioural Assessments (Retained)	The platform shall retain existing MCQ generation and behavioural assessment modules, with integration to the AI Fit Score pipeline.	P1	Candidate, Recruiter	Assessment results are stored and reflected in the candidate's composite AI Fit Score.
FR-028	JWT/RBAC Security (Retained & Enhanced)	All platform routes shall remain secured by JWT authentication and RBAC, with role definitions expanded to include new stakeholder types.	P1	All	Unauthorised access attempts return HTTP 403; all user actions are logged with timestamps.
FR-029	Audit Logs (Retained & Enhanced)	Comprehensive audit logs shall capture all user actions, AI scoring events, data access, and system errors with immutable timestamps.	P1	Auditor, HR Admin	Audit logs are queryable by date, user, action type, and entity; logs are retained for a minimum of 24 months.

8. Non-Functional Requirements

ID	Category	Requirement	Target Metric / Notes
NFR-001	Scalability	The platform architecture shall support horizontal scaling of individual microservices independently using Docker containers orchestrated via Kubernetes or AWS ECS.	Each service auto-scales independently; scaling events triggered at ≥70% CPU or memory utilisation
NFR-002	Scalability	The vector database (Pinecone/Weaviate/FAISS) shall support indexing and querying of up to 10 million resume vectors without performance degradation.	Query latency ≤ 500ms at 10M vectors under standard load
NFR-003	Performance	AI Fit Score computation for a single candidate-job pair shall complete within an acceptable response time under normal load.	≤ 5 seconds per single scoring request; ≤ 60 seconds for batch of 500
NFR-004	Performance	Semantic search queries via vector similarity shall return results within acceptable latency bounds.	≤ 3 seconds per natural language search query
NFR-005	Reliability	The LLM Gateway shall implement multi-model routing with automatic fallback to secondary LLM providers if the primary provider is unavailable or exceeds latency thresholds.	Fallback triggered if primary LLM latency > 10s or error rate > 5%; fallback completes within 2s
NFR-006	Reliability	The system shall implement circuit breakers and request throttling to prevent cascading failures across AI services.	Circuit breaker opens after 5 consecutive failures; throttling limits to 100 req/min per service
NFR-007	Availability	The platform shall maintain high availability during business hours and scheduled maintenance windows.	≥ 99.5% monthly uptime SLA; planned maintenance notified 48 hours in advance
NFR-008	Security	All data in transit shall be encrypted using TLS 1.2 or higher; all data at rest (AWS S3, MongoDB) shall be encrypted using AES-256.	Zero unencrypted data transmission; encryption enforced at infrastructure level

NFR-009	Security	JWT tokens shall have a configurable expiry (default 8 hours) and RBAC policies shall be enforced at every API endpoint.	Expired tokens return HTTP 401; RBAC violations return HTTP 403; all enforced at middleware layer
NFR-010	Observability	The system shall provide real-time monitoring of all AI service calls, including latency, error rates, and fallback incidents via AWS CloudWatch and Grafana dashboards.	Dashboards updated every 60 seconds; alert thresholds configured for latency > 5s and error rate > 2%
NFR-011	Maintainability	All services shall be containerised (Docker) with documented APIs (OpenAPI 3.0 specs) to ensure team independence and ease of maintenance.	All services have published OpenAPI specs; CI/CD pipelines in place for all microservices
NFR-012	Data Retention	Candidate data, audit logs, and AI scoring records shall be retained for a minimum of 24 months to support compliance and audit requirements.	Data retention policy enforced at database level; automated archival of records older than 24 months

9. Integration Requirements

Integration	System / Provider	Purpose	Protocol / Method	Priority
Cloud File Storage	AWS S3	Secure storage of resumes, documents, and media assets	REST API / AWS SDK	P1
LLM — Primary	GROQ API / GPT-4 (OpenAI)	AI Fit Scoring, content generation, candidate summaries, JD generation, email drafts	HTTPS REST / OpenAI SDK	P1
LLM — Fallback	Anthropic Claude	Secondary LLM for routing fallback when primary LLM is slow or unavailable	HTTPS REST / Anthropic SDK	P1
Vector Database	Pinecone / Weaviate / FAISS	Storage and cosine similarity	REST API / gRPC	P1

		querying of resume and JD embeddings		
Embedding Models	BERT / OpenAI Embeddings	Generating high-dimensional vector representations of text documents	Python SDK / REST API	P1
NLP Processing	spaCy (Python)	Named entity recognition: extract names, titles, skills, and certifications from CVs	Python library / microservice	P1
Email Delivery	Nodemailer (SMTP)	Transactional email delivery: invitations, rejections, reminders, notifications	SMTP / Nodemailer SDK	P1
Containerisation	Docker + Kubernetes / AWS ECS	Containerised microservice deployment and auto-scaling orchestration	Docker CLI / Kubernetes API	P1
Monitoring	AWS CloudWatch / Grafana	Real-time observability: latency, error rates, AI call health, pipeline monitoring	CloudWatch SDK / Grafana API	P2
Salary Market Data	External Salary API (TBD)	Market-based salary benchmarking and competitive pay range recommendations	REST API (provider TBD)	P2
Frontend UI	React / Vue.js	Recruiter dashboards, candidate pipeline views, chat UI, and analytics visualisations	REST API / WebSocket	P2
File Upload Handling	Multer (Node.js)	Multipart/form-data handling for resume and document upload endpoints	Express.js middleware	P1

10. Analytics & Reporting Requirements

HireGen AI shall provide comprehensive analytics and reporting capabilities across three tiers: operational recruitment metrics, AI performance metrics, and system health monitoring.

10.1 Recruitment KPIs (Dashboard)

KPI	Definition	Visualisation Type
Time-to-Hire	Average number of days from job posting to accepted offer	Line chart / trend
Offer Acceptance Rate	Percentage of extended offers accepted by candidates	Gauge / percentage
Pipeline Volume	Count of candidates at each stage: Applied, Screened, Interviewed, Offered, Hired	Funnel / bar chart
Open Jobs	Number of active, unfilled job postings	Number card
Applicants per Role	Total applicants received per job posting	Bar chart
Average AI Fit Score	Mean AI Fit Score across all applicants for a job	Distribution histogram
Resume Authenticity Rate	Percentage of submitted resumes flagged as Suspicious or High Risk	Donut chart
Top Skills in Demand	Most frequently requested skills across active JDs	Horizontal bar chart

10.2 AI Performance Metrics

Metric	Definition	Target
NDCG (Search Quality)	Normalised Discounted Cumulative Gain — measures search result ranking quality	≥ 0.85
AI Fit Score Accuracy	Correlation between AI Fit Score and recruiter final selection decision	≥ 80% correlation
LLM Latency	Average response time per LLM API call (scoring, generation)	≤ 5 seconds
Fallback Incident Rate	Percentage of LLM requests routed to secondary/fallback model	≤ 2% under normal load
Error Rate	Percentage of AI API calls returning errors or timeouts	≤ 1% under normal load

11. Compliance & Audit Requirements

Compliance Area	Requirement	Implementation Mechanism	Status
Hiring Decision Audit Trail	All AI scoring events, recruiter decisions, and status changes must be logged with actor, timestamp, and action.	MongoDB audit log collection with immutable write policy	In Scope — Phase 1
Explainable AI Compliance	Every AI Fit Score must be accompanied by a documented justification (XAI report) accessible to recruiters and auditors.	XAI justification panel + downloadable compliance reports	In Scope — Phase 1
Role-Based Data Access	Candidate personal data (PII) shall only be accessible to users with appropriate RBAC roles.	JWT + RBAC middleware enforced at all API endpoints	In Scope — Phase 1
Bias Mitigation	AI-generated job descriptions shall be reviewed for exclusionary language, with flagged terms highlighted and alternatives suggested.	LLM-powered bias detection in JD Generator service	In Scope — Phase 2
Resume Fraud Detection	Submitted resumes shall be assessed for AI-generated or falsified content, particularly for compliance-sensitive roles (finance, healthcare).	Resume Authenticity Score engine with red-flag detection	In Scope — Phase 3
Data Retention Policy	Candidate data, assessment records, and audit logs shall be retained for a minimum of 24 months.	Database-level retention policy with automated archival	In Scope — Phase 1
GDPR / SOC2 / EEOC	Data privacy (GDPR), security certification (SOC2), and equal employment compliance (EEOC) frameworks.	Formal compliance assessment and certification engagement required	Recommended — Out of Scope v1.0

12. Implementation Roadmap

The HireGen AI platform enhancement is planned across four phased delivery milestones. Phases are prioritised by business impact, technical dependency, and implementation effort. Each phase delivers a set of production-deployable features.

Phase	Feature	FRs Covered	Priority	Impact	Effort
Phase 1 (Months 1–3)	Core AI Matching & Scoring (Vector Search + Fit Score Engine)	FR-001–004, FR-008–011	P1	High	Medium
Phase 1 (Months 1–3)	Explainability & AI Summaries (XAI Panel + Profile Summaries)	FR-005–007, FR-012	P1	High	Low
Phase 1 (Months 1–3)	Recruitment KPI Dashboard & Visual Pipeline	FR-022, FR-023	P1	High	Low
Phase 1 (Months 1–3)	Resume Parsing Enhancement (spaCy NER Integration)	FR-026	P1	High	Low
Phase 2 (Months 4–5)	Interview Question Generation & Feedback Scoring	FR-014–016	P1	Medium	Medium
Phase 2 (Months 4–5)	AI JD Generator & Email Communication Automation	FR-020, FR-021	P1	Medium	Medium
Phase 2 (Months 4–5)	Salary Benchmarking Integration	FR-024	P2	Medium	Medium
Phase 3 (Months 6–8)	Semantic Chat Search (Natural Language Recruiter Assistant)	FR-013	P2	Medium	High
Phase 3 (Months 6–8)	Risk / Fraud & Authenticity Detection	FR-017–019	P2	High	High
Phase 3 (Months 6–8)	Skill Demand Trend Analytics	FR-025	P3	Medium	Low

Phase 4 (Months 9–12)	Full RAG-Based Recruiter Chatbot	FR-013 (extended)	P3	Medium	High
Phase 4 (Months 9–12)	GDPR / SOC2 / EEOC Compliance Certification Preparation	Compliance Section	P2	High	High

13. Assumptions & Constraints

13.1 Assumptions

- The existing Node.js/Express/MongoDB codebase is stable, well-documented, and accessible to the development team.
- GROQ API, OpenAI, and Anthropic Claude APIs will maintain their current pricing tiers and rate limits throughout Phase 1–2 development.
- Recruiters and HR Admins will participate in user acceptance testing (UAT) sessions for all Phase 1 features.
- AWS infrastructure (S3, CloudWatch) is already provisioned and accessible; additional AWS services (ECS, Lambda) can be enabled within the existing AWS account.
- A dedicated vector database instance (Pinecone or Weaviate) will be provisioned before the start of Phase 1.
- External salary data API integration is dependent on vendor contract execution; salary benchmarking feature is contingent on this.
- All resumes submitted to the platform are in PDF format; support for DOCX and other formats is a Phase 5+ consideration.
- The development team has proficiency in Python (for spaCy/NLP service) in addition to Node.js.
- Candidate data ingested during testing will use anonymised or synthetic datasets compliant with applicable privacy laws.

13.2 Constraints

- LLM API costs must be managed within the allocated monthly AI infrastructure budget; model routing (GPT-4 vs GPT-3.5) will be used to optimise cost-vs-quality tradeoffs.
- The EJS templating frontend will be progressively replaced with React/Vue; full frontend migration is not within the Phase 1–2 scope.
- All PII (personally identifiable information) must remain within defined data residency regions; no cross-border transfer without legal review.
- Vector database selection (Pinecone vs Weaviate vs FAISS) must be finalised before Phase 1 development begins; migration between providers is costly.
- The platform must remain fully operational (zero downtime) during the microservices migration; a blue-green or strangler fig migration pattern is required.

- Third-party LinkedIn API access for resume-LinkedIn consistency checks (FR-019) is subject to LinkedIn's API access policies and may not be available at launch.
- GDPR, SOC2, and EEOC compliance certification are deferred to post-Phase 4 and require dedicated compliance engineering and legal resources.

14. Risk Register

Risk ID	Risk Description	Category	Likelihood	Impact	Mitigation Strategy
RISK-001	Primary LLM provider (GPT-4/GROQ) outage or rate-limit throttling disrupts AI services	Technical	Medium	High	Implement LLM Gateway with multi-model routing and automatic fallback to Claude/GPT-3.5; circuit breakers prevent cascading failures
RISK-002	Vector database latency degrades under high query volume	Performance	Medium	High	Benchmark Pinecone vs Weaviate vs FAISS under load before production selection; implement query caching for frequent searches
RISK-003	AI Fit Scores exhibit bias against protected candidate characteristics	Compliance / Ethical	Low	Critical	Implement bias auditing on training data and scoring outputs; engage an independent AI ethics review; XAI justifications expose scoring drivers for manual review
RISK-004	Resume fraud detection produces high false-positive rate, flagging legitimate candidates	Quality	Medium	Medium	Implement confidence thresholds; all flags are advisory (not auto-reject); recruiter override mechanism required
RISK-005	Microservices migration causes regression in existing	Technical	Medium	High	Use strangler fig migration pattern; maintain existing monolith in parallel;

	workflows				comprehensive regression testing before each service migration
RISK-006	External salary API vendor fails to deliver data or contract falls through	Vendor	Low	Medium	Identify two alternative salary data providers; design salary benchmarking service with pluggable data source interface
RISK-007	LLM-generated JDs or emails contain inaccurate or inappropriate content	Quality	Medium	Medium	All AI-generated content is editable before publishing/sending; human-in-the-loop review step enforced for JDs and bulk emails
RISK-008	LinkedIn API access for resume cross-verification denied or restricted	Vendor / Regulatory	High	Low	FR-019 (LinkedIn consistency check) is P3; design as optional enhancement; proceed with internal authenticity scoring as primary fraud detection
RISK-009	GDPR non-compliance due to PII stored in vector embeddings	Regulatory	Medium	High	Engage legal/DPO review before Phase 1 deployment; implement data anonymisation pipeline before vectorisation; plan GDPR compliance sprint post-Phase 4

15. Glossary

Term	Definition
AI Fit Score	A composite numerical score (0–100) computed by HireGen AI that quantifies how well a candidate's profile matches a specific job description based on skills, experience, and education vectors.

ATS	Applicant Tracking System — software used to manage, track, and automate the recruitment process from job posting to hire.
BERT	Bidirectional Encoder Representations from Transformers — a transformer-based NLP model used to generate contextual text embeddings for semantic search.
Circuit Breaker	A software design pattern that prevents an application from repeatedly attempting to execute an operation likely to fail, protecting system stability.
Cosine Similarity	A mathematical metric used to measure the similarity between two vectors; central to semantic matching in the HireGen AI vector search engine.
FAISS	Facebook AI Similarity Search — an open-source library for efficient similarity search and clustering of dense vectors.
GROQ API	An LLM inference API used in the current HireGen AI stack for AI content generation tasks.
JWT	JSON Web Token — a compact, self-contained token used for secure user authentication and session management.
LLM	Large Language Model — a deep learning model trained on large text corpora capable of generating, summarising, and analysing natural language.
LLM Gateway	A centralised architectural layer in HireGen AI that orchestrates all LLM API calls, implementing routing, fallbacks, circuit breakers, and throttling.
NDCG	Normalised Discounted Cumulative Gain — a metric used to evaluate the quality of ranked search results by measuring how relevant the top results are.
NER	Named Entity Recognition — an NLP task that identifies and classifies named entities (e.g., person names, job titles, skills) in unstructured text.
Pinecone	A managed vector database service optimised for similarity search at scale, used for storing and querying resume and JD embeddings.
RAG	Retrieval-Augmented Generation — a technique combining vector search retrieval with LLM generation to produce contextually accurate AI outputs.
RBAC	Role-Based Access Control — a security model that restricts system access based on predefined user roles and permissions.

Semantic Search	A search methodology that understands the meaning and context of a query rather than relying on exact keyword matches.
spaCy	An open-source NLP library in Python used for advanced text processing including NER, tokenisation, and dependency parsing.
Vector Embedding	A numerical representation of text as a high-dimensional vector that captures semantic meaning, enabling similarity-based comparisons.
Weaviate	An open-source vector database that supports semantic search and is used as an alternative to Pinecone in the HireGen AI architecture.
XAI	Explainable Artificial Intelligence — AI systems designed to provide human-understandable explanations for their outputs and decisions.
